

Research article

TSM: Temporal segmentation and modules-based computation offloading using predictive analytics and NR-V2X[☆]

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ABSTRACT

Recent increases in the usage of applications dependent on distributed computing have persuaded designers and entrepreneurs to utilize these solutions for diverse purposes, especially latency-constrained computation-intensive applications. Vehicular fog computing (VFC) is the innovative paradigm of distributed computing techniques, and therefore, a number of VFC offloading frameworks have been developed using AI-based advanced optimization procedures, with support from standards maintenance organizations like IEEE, 3rd Generation Partnership Project (3GPP), and some others. However, many of these strategies have not been adapted to specific application data and types, therefore these frameworks may function poorly despite their comprehensive offloading principles. Moreover, most computation offloading frameworks ignore the use of updated V2X protocols. We designed a temporal segmentation and modules (TSM)-based method specific for computation-intensive V2X applications that uses a four-tier hierarchy of resource-rich nodes and works in discrete time periods. TSM relies on status updates from previous time periods and uses predictive analytics to address the stochastic nature of vehicular networks using the latest 3GPP 5G V2X standards. Using an online modular computation offloading structure that heuristically manages the whole process, we were able to successfully and timely execute the latency-sensitive advanced vehicular applications. TSM supports computation-deficient devices in under a hundred milliseconds, makes use of smart vehicles' processing units as fog nodes, and solves the optimization problem in short, discrete stages. We utilized Monte Carlo analysis, which confirmed that TSM outperformed the three other baseline methods.

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1. Introduction

Our current era of intelligent systems and environments—such as autonomous driving [1], smart cities, smart transportation [2], augmented and virtual reality [3], interactive gaming, video streaming, face recognition applications, healthcare (health 4.0) [4], intelligent [5] and big data analytics [6] and several similar-generates massive amounts of data that needs to be processed in real-time. According to recent statistics from the International data corporation (IDC), 73.1 ZB of data would be produced by the 55.7 billion devices performing such real-time services in 2025. Real-time applications need large amounts of data to be processed rapidly and accurately, yet these devices have limited hardware and software resources. As a result, these devices often use distributed computing solutions for computation-intensive applications [7]. Cloud, edge, and fog computing are well-known and widely used paradigms that define distributed computing infrastructures to compensate for the lack of computing resources [8]. Users may remotely access virtualized computer infrastructure, platforms, and software services without managing the underlying infrastructure using cloud servers. However, cloud servers, being at relatively far places, needs more compensation from alternative options like edge and fog computing. Edge and fog computing are two terms that are often interchangeably used and carry a very vague distinction in their utility and meaning [9]. For example, one of the main differences is the relatively larger distance of an edge server from a needy node as compared to that of a fog node from the needy node. To give a further comprehensive description of a special type of distributed computing solution, the Vehicular Fog Computing (VFC)-enabled scenarios, we can consider the on-board computer unit (OBCU) of its member smart automobiles as fog nodes, while the Road Side Units (RSUs) as edge nodes, also known as a vehicular edge Computing (VEC) servers [10]. The VFC or VEC is fundamentally reliant on strong networking infrastructure, which is provided by the well-known V2X paradigm, which is, in turn, reliant on Collaborative Autonomous Vehicles (CAVs) or Road Side Units (RSUs) for handling distributed computing applications. Other key significant factors in improving the overall efficiency of a fog computing-enabled system are coherence and interoperability among participating devices (nodes). Given these specifications, the following are some of the most common challenging circumstances:

- Given that autonomous vehicles and other computation and data-intensive applications regularly generate vast amounts of data in their surroundings, how this data may be efficiently maintained, communicated, and processed?
- How can the high cost of computing demands be fulfilled, given that the computational capabilities of many neighboring devices may not, on their own, always meet the ever-increasing computing demands?

The aforementioned challenges have something in common: both involve time limits and require additional computing power. A good task-offloading strategy provides workable answers to the above-identified problems.

1.1. Relevant case studies and possible application scenarios

Besides introducing an efficient offloading strategy, identifying possible application scenarios is also one of the important steps for developing a viable VFC-enabled distributed computing framework. For this purpose, with the aid of various applicable case studies, we were able to verify our approach and the possible outcomes. In this regard, the previously published papers as outlined in the “related work” Section 2 allowed us to identify several applicable use cases; we used this information as the foundation for evaluating our approach by choosing an appropriate case study to serve as a potential application scenario. Various published papers’ descriptions of real-world deployments of edge computing led us to affirm that the 3rd Generation Partnership Project (3GPP) is an excellent resource to find such instances of application. For this reason, as can be seen in Fig. 1, we chose 3GPP-endorsed “smart cities” as one of the “possible application scenarios” for our proposed study.

Another related factor is the role played by edge or fog computing in ‘smart cities’ that can be further categorized into several types, such as traffic management and optimization, public safety and security, smart street lighting and energy management, waste management, environmental monitoring and air quality control, smart parking, public transportation optimization, healthcare and emergency services, water management, tourism, and cultural services. However, we considered our ‘possible application scenarios’, those work areas of ‘smart cities’ that require edge or fog computing assistance and can operate consistently with a latency of a few milliseconds to two hundred (200) milliseconds.

1.2. Contributions

In this article, we analyze the latency reduction problem of resource-constrained nodes, which can be smart vehicles, their passengers, and also pedestrians, to ensure a timely task offloading process. The tasks are uniquely created by different applications and have various requirements for CPU cycles and task data size. In addition, the transmission rate fluctuates intermittently throughout the period since the vehicles are in constant motion. As a consequence of these variables, the computational load, scheduling, and final allocation requirements of a given vehicular operation are always shifting. Due to the constantly shifting dynamics of the scenario, developing an effective offloading strategy is challenging. We present a probability distribution function (PDF)-based analysis of our method as a performance indicator, which verifies the minimum time consumed throughout the offloading process that is in turn based on predictive analytics. This article makes the following major contributions:

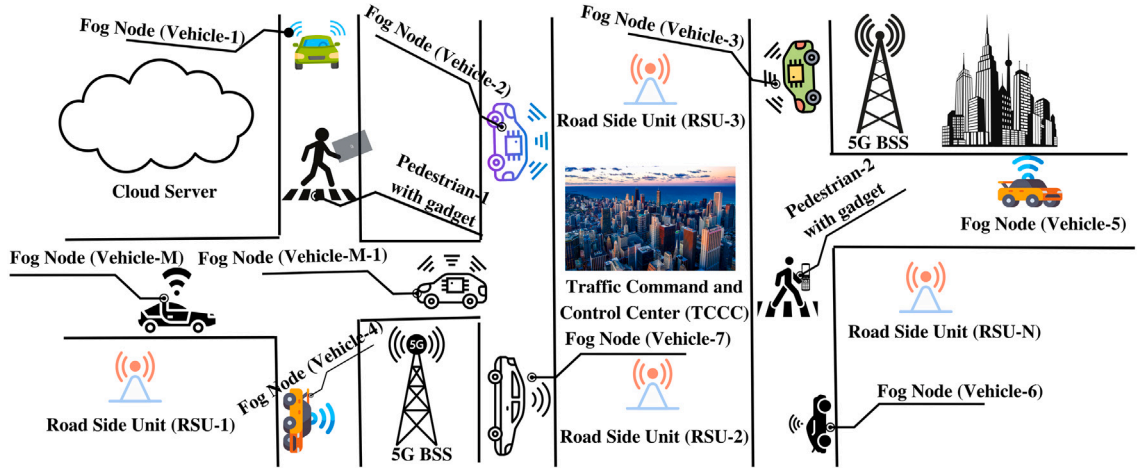


Fig. 1. VFC application scenario.

Table 1

List of V2X-specific protocols and their attributes in context of distributed computing.

Protocol Name	ITS-G5	DSRC/IEEE 802.11p	3GPP LTE-V2X	3GPP NR-V2X	IEEE 802.11bd
Radio bands	5.875 GHz–5925 GHz [12]	5.895 GHz–5.925 GHz [13]	5.855 GHz–5.935 GHz [14]	FR1 (2.570 GHz–2.620 GHz, 5855 GHz–5925 GHz) and FR2(24.25 GHz to 52.6 GHz) [15]	5.9 GHz, and 60 GHz [16]
Preferred applications area	Safety, Non-safety, and ITS-applications [12]	Safety related [17]	Advertising [18] and non-safety [17]	Advanced vehicular, and vertical applications for spectrum/network bandwidth, storage, and computation [19]	Advanced vehicular [16].
Latency Support	≤ 100 ms [20]	150 ms [17] and 100 ms [21]	≤ 10 ms [19] and 100 ms [16]	≤ 1 ms [16]	100 ms [17]
Data rate	3–27 Mbps & 6–54 Mbps [22]	3–27 Mbps	26 Mbps TX/RX [23]	25 Mbit/s and 1 Gbit/s [24]	54 Mbps [25]
Mobility support	110 Km/hr	120 Km/hr	120 Km/hr	500 Km/hr	500 Km/hr [26]
Suitability for type of traffic	Urban, Rural, Suburban	Urban	Rural	Rural and Urban both	Urban
Network Slicing	No	No	No	Yes	Yes

- We suggest a two-tiered VFC architecture, whereby certain vehicles, pedestrians, and passengers act as needy nodes by generating tasks, while four different types of computation resource-rich nodes with ample computational capacity act as donor nodes. Furthermore, infrastructures like RSU and neighborhood smart vehicles are considered stationary and mobile fog nodes. Stationary fog nodes are stationary computing resources in the vehicle environment. Roadside infrastructure, access points, and other computing devices along the road network are these nodes. Stationary fog nodes analyze data and offer services near vehicles using computation and storage [11]. They may use wired or wireless network connections. Stationary fog nodes handle vehicle data at important locations using fixed computing resources. Conversely, mobile fog nodes are vehicular computing resources. These nodes are usually onboard computers or fog computing devices in vehicles. They may help process data near the source, decreasing latency and bandwidth. For purposes of work distribution within its own sphere of influence, we considered each fog node as a potential controller with the ability to schedule and allocate resources.
- With timely task scheduling and resource allocation and coherence among different operating modules of TSM in focus, this article aims to optimize task offloading and resource allocation to boost computational resource accessibility to multiple types of consumers using a separate classification based on existing V2X paradigms such as Vehicle to Infrastructure (V2I), vehicle to vehicle (V2V), Vehicle to Network (V2N), Vehicle to Pedestrian (V2P), and Pedestrian to Infrastructure (P2I). The timely offloading procedure is successfully confirmed by the response time, which is attained utilizing a distributed technique and measured in milliseconds.
- For our approach, the TSM framework of distributed scheduling and allocation methods is supported by heuristically solving its optimization problem. It repeatedly moves through the resource allocation and offloading choices until the appropriate decisions are reached. We do in-depth simulations to assess and ensure the suggested strategy. The experimental results show that the proposed method successfully completes the offloading process while fulfilling task computation time for a variety of latency-sensitive computationally-heavy applications with unique latency requirements, as compared with benchmark techniques.

- We followed the norm that VFC-enabled systems, like any other, should follow relevant standards issued by an organization actively involved in standard creation and development to ensure network component consistency and interoperability [27]. For this purpose, we relied on 3GPP 5G NR, based on which we were able to finalize the inter-networking of participating devices (and nodes) in the VFC-enabled environment.

1.3. Paper organization

This article's remaining sections are organized as follows. The system model is presented in Section 3, followed by a discussion of the problem formulation in Section 4. Section 5 provides a hybrid approach of distributed and centralized methods and a strategy for offloading and allocating tasks based on latency constraints, taking mobility into account. Section 5.3 contains the numerical outcomes. The article is concluded in Section 6.

2. Related work

Knowing the adaptability of CAVs to distributed computing, a number of computation offloading approaches, have widely utilized IEEE 802.11p and 3rd Generation Partnership Project (3GPP) LTE-V2X Side Link (SL) protocols. This is because both of these standards, which were developed in 2010 and 2016, have been in effect for a longer period of time. For a comparison of currently available and actively developed V2X protocols, the Table 1 gives us a good starting point. In terms of how these protocols are used, for example, the article cited as [28] evaluated IEEE 802.11p by considering OBCUs of approaching vehicles as fog nodes. The suggested fog computing infrastructure's average end-to-end latency by scheduled computing tasks is between 40 and 62 ms [28]. A recently proposed computation offloading architecture based on NR-V2X, such as [9], schedules, allocates, and offloads tasks with a delay of 0.4 to 1 s, which is far higher than the delay exhibited in [28]. However, when exposed to heavy traffic, high vehicle speeds, and a broad diversity of locations, the computation offloading structure in [28] badly affected with higher delays. This is because IEEE 802.11p has reduced spectrum efficiency, and therefore, it is less suitable for many delay-sensitive applications, particularly autonomous vehicular applications.

Another fairly optimized computation task offloading framework based on the 3GPP 5G NR Uu and PC5 interfaces is in the article cited as [29]. The 5G Uu interface is the wireless link between a user equipment (UE) such as a smartphone and a base station (gNodeB) in a 5G network. The 5G network design relies on it to transmit data, control signals, and other information between the UE and the network. Whereas the other type, which is the PC5 interface, enables direct device-to-device (D2D) communication. The PC5 interface, unlike the Uu interface, lets neighboring devices interact directly without a base station. The 5G PC5 interface offers a direct communication route for adjacent devices, enabling efficient and low-latency data exchange in proximity for different 5G use cases. The offloading framework proposed in the article cited as [29] uses both the PC5 and 5G Uu interfaces, but unlike our approach, they only use Vehicle to Vehicle (V2V) interfaces, which weakens their proposal because several static donors and needy nodes are not considered. Their reliance on the clustering of smart vehicles into donor and needy nodes using the PC5 interface and another group using the Uu interface is another abstraction worthy of consideration. This abstraction could further be improved by classifying their framework into paradigms like Vehicle to Network (V2N) and Vehicle to Infrastructure (V2I) which would have made their categorization into various groupings more comprehensive.

Article cited as [9] addressed energy conservation and latency reduction problems in a Vehicular Edge Computing (VEC)-enabled scenario by proposing a joint computation offloading framework. Although they have not explicitly mentioned the host controller node for their proposed framework, we inferred that their model requires a centralized approach to solve the two most vital requirements. In contrast to our approach, the offloading framework proposed in [9] will be utilized by a centralized controller, like a roadside unit (RSU) or a base station (BS). Among the pros and cons associated with centralized scheduling and task allocation as compared with a distributed version is the trade-off between the amount of overhead due to control information dissemination and end-to-end latency reduction. And in the race to achieve a favorable balance in such a trade-off, the distributed approach is the winner [26,30].

A workload distribution in a dynamic vehicular environment is fairly balanced using another centrally managed computation offloading framework in the article cited as [11]. The framework ultimately depends on factors like the service rate of tasks, queuing theory, vehicle arrival rate, dwelling time of vehicles, and optimal allocation probability. However, factors like calculating probability accuracy, aided by overhead costs to timely provide those metrics based on which Mobile Edge Computing (MEC) server calculates the probabilities in a centralized task scheduling and allocation scenario, are prone to a significant number of errors. Let us reiterate the metrics used in their framework in [11] here, namely, mobility features of vehicles (i.e., staying time and turning probability), vehicle density patterns (i.e., vehicle arrival rate), and computation capabilities of MEC/cloud servers (i.e., the variety of processors), in order to get an idea of the significant margin of error due to overhead costs associated with broadcasting and transmitting these metrics to a centralized scheduler.

Article [31] references a recently published compute offloading framework intended to meet the critical needs of real-time applications in autonomous driving, such as positioning and obstacle prevention, using the deep reinforcement learning (DRL) method. Even though MEC-augmented high-definition (HD) map generation for localization in latency-sensitive scenarios is a key requirement for new V2X scenarios, the DRL-based approach can be replaced by relatively less computationally intensive approaches [32]. Classical machine learning algorithms, feature selection, dimensionality reduction, incremental or online learning, and knowledge distillation (teacher and student models) are some of the alternative methods options that have relatively less computational overhead [33].

Another computation offloading framework that formalizes the vehicular fog computing architecture is presented in the article referenced as [34]. It focuses on conflict situations (such as urban combat and battlefields in the Internet of Battlefield Things) and employs a fog-assisted traffic management system as a use case. A strong argument that strengthens the validity of their fog-computing architecture is their targeting of a particular set of applications, which are vehicular applications where specifying a particular set of applications makes it easy to design an efficient architecture. In addition to the application-level detail, the overall architecture is abstracted into granular sublayers, making it easy to understand for other developers. While the concept recognizes the requirement for communication between nearby vehicles (such as fog nodes or the cloud server), it does not provide details of the usage of any particular low-cost and reliable Local Area Network (LAN) or City Area Network (CAN) for such purposes.

Time-varying spectrum efficiency (SE) owing to time-varying fading channels and additional uncertainly allotted bandwidth was addressed effectively in one other task offloading framework in article [35]’s utility maximization problem with task delay requirement constraints. In addition, the suggested non-convex optimization problem has been efficiently solved using linearization-based branch and bound (LBB) and closest rounding integer (CRI) techniques. However, their computation offloading model relies on a single VEC server. The extra flow of data due to disseminating control data and status updates may be mitigated by relying on a centralized VEC server. The authors’ suggested offloading technique, however, does not make use of the simplicity and comprehension provided by the abstraction of employing several V2X paradigms, such as V2V, Vehicle-to-Infrastructure (V2I), and Vehicle-to-Network (V2N). Another issue is that they only evaluate one type of wireless channel between automobiles and infrastructure (the roadside unit (RSU) or Vehicular Edge Computing (VEC) server), while in fact many alternative paradigms may have been explored had they been brought up.

Joint roadside unit (RSU) selection and resource allocation problem, with the objective of minimizing the total task offloading delay subject to bandwidth and computation resource constraints using collaborative and non-collaborative RSU-based modeling, is another interesting computation offloading framework in the article cited as [36]. The authors effectively used techniques like the alternating direction method of multipliers (ADMM), linearization, and branch and bound (LBB) algorithms to first transform and then solve the non-convex problems which are subsequently based on non-collaborative, and collaborative scenarios. It is important to mention that the use of IEEE 802.11bd as in Table 1, which is the latest update to the predecessor IEEE 802.11p protocol, would have brought significant improvements to their proposal in the mentioned article [36], especially in terms of spectral efficiencies. However, the use of carrier-sense multiple access and collision avoidance (CSMA/CA) of IEEE 802.11p in their communication model enhances the system’s utility, but the relatively limited ability of IEEE 802.11p to effectively ensure good communication in a relatively wider physical region dampens the overall strength and efficiency of their proposal.

Not all previous studies used the abstraction of the well-known V2X paradigms, including V2V, V2I, V2N, and vehicle-to-pedestrian (V2P), for defining their vehicular fog computing (VFC) architectures for task scheduling and allocation, with the purpose of optimizing strategies to maximize task offloading or computation resource allocation. It is because several offloading frameworks have not explicitly identified passengers and pedestrians as possible key consumers of the offloading processing, disregarding crucial abstractions like V2P. Also, several of the proposals, like in [28,36], have used older IEEE protocols for the networking of smart vehicles. Additionally, task scheduling and allocation are both important steps in a distributed computing process. However, these are driven by one of the main factors, which is whether they are distributed or centralized, a factor that is not given due consideration by many offloading frameworks. Though it is a fact that a distributed approach to implementing scheduling and allocation can improve efficiency over a purely centralized approach, at the cost of some overhead in the form of increased control and status updates information and computation specific for additional management and control [7,26]. In this context, our distributed scheduling and allocation approach with multiple potential controller nodes, helped us address efficiently resource-constrained and latency-sensitive applications. In contrast to existing offloading methods for VFC, our system takes into account the abstraction of tasks into relevant V2X paradigms as shown in the Ishikawa (fishbone) diagram in Fig. 4, and it is supported by decentralized scheduling and allocation. We have avoided placing Fig. 4 here and instead have shown it near Section 3.1.2, as more relevant details regarding V2X paradigms using Fig. 4 are explained in Section 3.1.2. Hence using V2X paradigms, we develop a latency-minimization-based computation-task offloading and resource allocation method that takes mobility into account, and we use a Monte Carlo simulation to validate the timely completion of the offloading process.

3. System model

We now present our model of VFC-enabled system which in turn is dependent on network topology, communication model, time response calculation, computation execution model, and waiting time model. The utility functions of the vehicle and VFC are then thoroughly discussed. All the notations that are used in this section are shown in Table 2.

3.1. Network topology

Fig. 1 illustrates our VFC network, which consists of “ V_N ” number of vehicles randomly distributed. Roads in an urban area are analyzed here, with initially a single roadside unit (RSU) installed along the roadside and an RSU server consisting of “ p_D ” compute power, as indicated in Table 2. The whole scenario encapsulates smart vehicles and their passengers, connectivity networks, cloud and edge servers (RSU), pedestrians, and road infrastructure. The vehicle set is represented by $V_n = 1, 2, 3, \dots, V_N$ and task requests by $Tks = 1, 2, 3, \dots, T_N$. Whereas “ Tks ” is the “id” of tasks being requested and “ T_N ” is the total number of tasks in a particular time period “ T ”.

Table 2
Purpose and values of input parameters.

Abbv.	Purpose	Values
V_S	Vehicles' average speed	[50–70] Km/h
ρ	Power by RSU	120 mW
R_N	Number of RSUs	1
$\hat{P}$	Transmission Power by vehicular fog node	80 mW
τ	Total number of time-instants in a specific period of time	100
T_H	Device response time in milliseconds as threshold (T_H) used in Eq. (11)	250
C_τ	Connection period between donor (SVs and RSUs) and needy nodes (TV/CUs) (in milli-seconds)	[1, 3.5]
T_N	Number of Tasks	[100, 300]
V_N	Vehicles' density in neighborhood	[50, 100]
$\hat{S}_z$	Size of a packet or a transport block (TB) in Kilobytes	[100, 1000] [37]
p_D	Computation Power at each smart vehicle [9]	[10, 15] GHz
CP_I	Computation Power at RSU and local data center	[15, 20] GHz
p_C	Cloud-server computation processing power	50 (Giga-Hz)
G_θ	Decision on offloading among fog vehicles, RSU, data center, and cloud-server	[1, T_N]
λ	Path Loss exponent [9]	3.2
χ	Computation processing power needed for each task	[50, 100]
SV_θ	Algorithm 2 Based 2nd Offloading decision for Fog Vehicles	[1, V_N]
B_C	Backhaul bandwidth from infrastructure to first terminal for Cloud-Server link.[38]	100
L_{BN}	Backhaul network propagation delay [39]	100 ms
$\hat{B}$	Needy node-infrastructure wireless channel bandwidth in MHz [40]	80
NR_I	Wireless interference and noise between needy node and infrastructure	−70 dB
NF_I	Wireless channel interference and noise between needy-fog vehicle and donor-fog vehicle	−75 dB

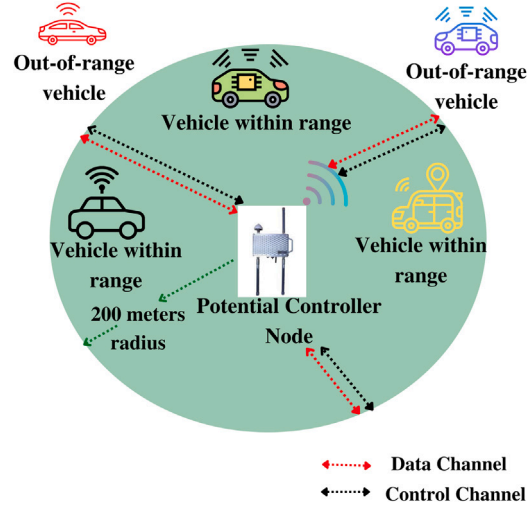


Fig. 2. Neighborhood criteria for declaring a smart vehicle as a “vehicular donor fog node”.

We used two generic terms to designate the needy node as resource-constrained and the donor node as a resource-rich neighborhood node. A pedestrian, a passenger, can only be a needy node. However, we considered autonomous smart vehicles as both needy and donor nodes, which we termed Task Vehicles (TV) and Service Vehicles (SV), respectively. The TV is a smart vehicle that is short of computation-task processing power and needs an external resource whereas the smart vehicles with redundant resources that can work as donor nodes are termed Service Vehicles (SVs) which we subjected to neighborhood criteria to be considered as vehicular fog node as shown in Fig. 2 and discussed in the section below. We adapted available computation resource parameter, p_D with values as in Table 2 and procedure adapted in Algorithm 1 to ensure adequate utilization of computation power of neighborhood smart vehicles. The donor nodes, assuming they are vehicular fog nodes, must meet neighborhood requirements. The needy nodes, which might be pedestrian or vehicular, are considered to exist within the neighborhood.

3.1.1. Neighborhood criteria

At this stage, it is critical to determine what factors constitute a neighborhood capable of maximizing resource-rich smart vehicle use, as indicated in Fig. 2. To solve this, first, we declared that the ‘potential controller node’ is the center of the surrounding area, as illustrated in Fig. 2, for our preferred neighborhood value. For this purpose, we copied a value of 200 m from one of the publications referred to as [24] that did a system-level study of mode-2 3GPP NR-V2X to define a smart vehicle as a donor vehicle. But whether or not the considered distance of 200 m is a suitable criterion to define the neighborhood of the controlling node? As a

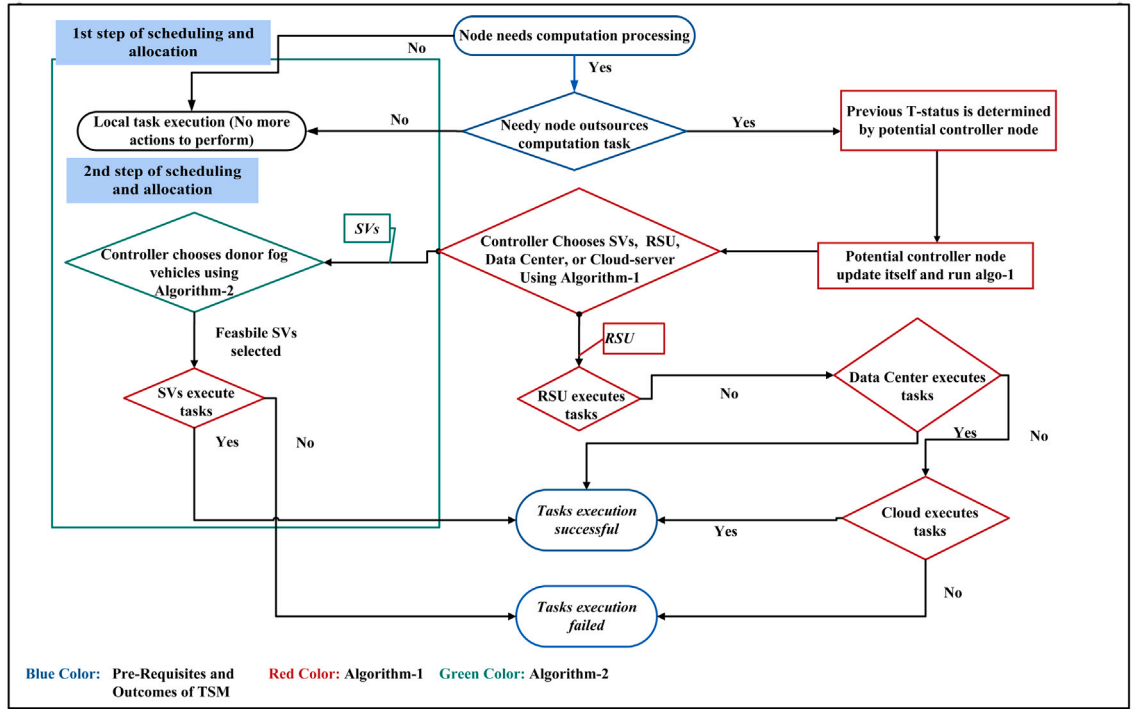


Fig. 3. Workflow of TSM.

work item (WI) of the 3GPP release-18, 3GPP is closing close on a solution to this problem by concluding the appropriate distance between User Equipment (UE) and their central controller [15,41]. Any network that adopts this proposed definition by 3GPP WI of the neighborhood will benefit from the optimized neighborhood made possible by this distance. Using our existing neighborhood criterion, we defined the network scenario using the following UE:

- SVs (Donor Fog Vehicles),
- Pedestrians or commuters with 5G subscription,
- Needy smart vehicles (TVs), and
- RSU
- Local data center

The overall structure of the VFC offloading process is depicted in Fig. 3. The ' G_θ ' and ' SV_ϕ ' are the two decision variables that are used by TSM in its Algorithms 1 and 2. However, both the pseudocodes are available in Section 5 because we have explained the entire approach of TSM distributed scheduling and allocation process in separate Section 5.

3.1.2. Mobility model

Our considered VFC network is based on the premise that every vehicle arrives at a random speed. The distribution of the vehicle speeds is similar and independent. In general, the speed of each vehicle, as it drives down the road, may vary over time. Therefore, each vehicle is assigned a random speed based on the values shown in Table 2, and each vehicle retains its assigned speed. Furthermore, it must be assured that the tasks are completed before the vehicles transition from the associated node to another node and the computation task delay is realized. This requires determining a vehicle's stay within a defined neighborhood of the controller, which is discussed earlier.

3.1.3. TSM V2X-networking model

Taking into account the V2V, V2I, V2P, and vehicle-to-cloud networks, we built a framework wherein task vehicles (TVs) and mobile phone users (CUs) could interact with sponsoring nodes (SVs), remote service units (RSUs), and cloud servers (Fig. 4). We use a specific diagram known as the Ishikawa (fishbone) diagram (Fig. 4) to illustrate the format in which our offloading approach can be depicted in terms of much talked about V2X paradigms. TSM uses the algorithms described to determine which of these paradigms best fits a given offloading process. Below is an outline of each of the auxiliary elements of the layout.

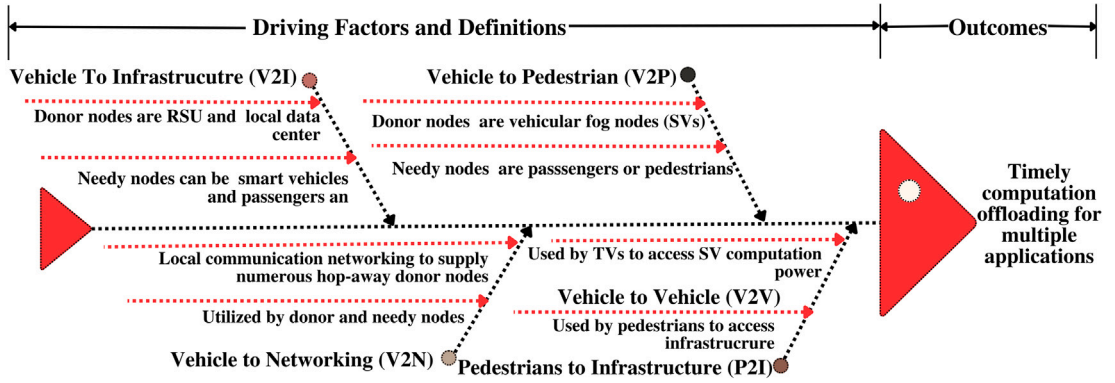


Fig. 4. Ishikawa (fishbone) diagram showcasing V2X-paradigms and our approach and their purpose.

V2I: We utilized V2I-abstraction to illustrate smart vehicles as recipients and also providers of computation resources. However, there can be additional donors of computation resources in this context and these are RSUs and Cloud-servers. Also, cloud servers and RSUs can be considered as the main elements building the infrastructure. The accessibility between the two mentioned ends is ensured using the 3GPP-endorsed 5G network. While needy nodes, as indicated in Section 3.3.2, if need a cloud server for computational tasks, may intrude on the backhaul network through 5G-based gNodeB and other terminals as shown in Fig. 1. Besides, the Roadside Unit (RSU) is an external resource and a potential controller node; hence, V2I is a good abstraction to comprehend this whole scenario.

V2V: This concept allows us to consider smart vehicles to communicate with one another regardless of whether donating or receiving the computation resources. With the NR-V2X guidelines' Side Link (SL), as detailed in Section 5.2, we were able to coordinate ad hoc wireless communication between vehicles whenever it was needed. According to Sections 5.2 and 5.1.1, NR-recommended V2X's scheduling plays a crucial role in maintaining a stable V2V network.

V2N: The main objective of the vehicle-to-network (V2N) paradigm is to classify communication between those donor and needy nodes which are smart vehicles but who are several hops away from each other. This is a powerful classification that helped us to outline one of the complex scenarios while handling the offloading process. Here an important observation is that a commuter's smart device may use the smart vehicle's 5G subscription instead of the commuter's own 5G subscription, thus increasing the number of interfaces for an onboard user to avail the high-speed and high-throughput connectivity.

V2P: We use the V2P classification for a pedestrian who may utilize computation assistance offered by donor fog vehicles, as shown in Fig. 4, despite the fact that the primary goal of V2P is to identify and communicate neighborhood pedestrian vulnerability [42]. We treated a pedestrian's smart device or mobile phone as a deficient node and a smart vehicle as a donor that may supplement pedestrians by the use of a vehicle-to-pedestrian (V2P) link. The donor smart vehicle in our architecture must, however, be a donor fog vehicle, as shown in Fig. 2, and conform to the neighborhood conditions outlined in Section 3.

Pedestrian to infrastructure (P2I): When employing interactive apps, there may be a scarcity of resources for users other than those traveling in smart vehicles who are using their own gadgets. Donor vehicular fog nodes may not always be readily available, therefore it may be necessary to rely on nearby infrastructure like an RSU, a local data center, or a cloud server to make up for the shortcomings. To better explain the process involved in developing our TSM algorithms, we found that the P2I paradigm was particularly helpful. Adding P2I to the well-known V2X paradigms currently in use is a significant development, as seen in Fig. 2. However, we have merely included P2I in the collection of principles to reinforce this classification.

3.2. Communication model

In V2I, V2N, our defined P2I, and V2V, V2P communication, vehicles, and pedestrians communicate with RSUs and local data centers and each other via 5G NR-V2X Mode-1 and Mode-2 wireless networks. We used two other classifications to better outline our proposal, one cellular link and another ad-hoc link. The cellular links contain V2I, V2N, and P2I links while the ad-hoc links contain V2V and V2P links. We assume that all vehicles and RSUs are equipped with several antennas for communicating over fifth-generation (5G) wireless networks. In addition, [43] suggests a distance ranging from 100 and 200 m between network nodes. Hence, we assumed that the V2V and V2P communication range is 150 m, while the V2I, V2N, and P2I communication ranges are 200 m. Following are the detailed explanations of the aforementioned links, determined from the intermediate steps we identified.

3.2.1. Cellular links

The cellular link falls under NR-V2X's Mode-1 in V2I communication [16]. With this fact, we considered other links with infrastructure as one of two ends, that are P2I and V2N to be handled via Mode-1 of NR-V2X. 3GPP Release 16 introduces the first V2X standard, which is based on 5G NR, which was published in 3GPP Release 15. Advanced V2X applications with more rigid QoS requirements than those supported by Cellular V2X can be handled by NR V2X as can be compared with other protocols shown in Table 1 [44]. 5G NR-V2X in the context of 3GPP guarantees enhanced latency, throughput, connectivity, mobility, and reliability. The implementation of the gNBs in this article is seen as a standalone node linked to the user plane function (UPF) and V2X application server, which are colocated with the pure 5G system core and access components. While the RSUs are presumptively implemented in accordance with 3GPP Release-16 NR-V2X specifications as standalone UE-type RSU node [15]. Along with RSU, we also considered local data centers as part of the infrastructure because they use similar cellular links and are colocated with a "gNB" while using Mode-1 recommendations of NR-V2X. Based on these considerations, we calculate the expected transmission data rate denoted as ' $\hat{R}$ ' between the needy node and infrastructure using the following mathematical expression.

$$\hat{R} = \hat{B} \log_2 \left(1 + \frac{\hat{P}[r(\hat{L}_n/r - \hat{L}_n)]^{-\lambda}|h^2|}{\hat{I}} \right), \quad (1)$$

While power transmission is defined as ' $\hat{P}$ ' between a donor fog vehicle (SV) and a needy node (CUs and TVs), ' $\hat{I}$ ' as interference plus noise, and ' $\hat{B}$ ' as bandwidth over a wireless channel between these nodes. ' $\hat{L}_n$ ' the vehicles' current location, ' r ' radius of neighborhood region with potential controller node as the center, ' λ ' the path loss exponent, and ' $|h^2|$ ' as Rayleigh fading channel also used in Eq. (2).

3.2.2. Adhoc links

The interactions that are defined by "V2V" and "V2P" abstractions do not depend on "gNB" for their task scheduling and allocation processes. Therefore, such connections can be well defined by the term "ad-hoc", as these connections are used when donor and needy nodes interact with each other without involving a "gNB", if the nodes are invoked for possible support, and if they are within the defined neighborhood region as depicted in Fig. 2. In terms of 3GPP NR-V2X, these interactions are endorsed and explained by Mode-2. These are the reasons that a different wireless channel communication and hence a unique mathematical expression for determining the transmission data rate between the vehicular fog node and the needy node as the ad-hoc link is used. For establishing a workable transmission data rate expression, we first explicitly create a signal-to-noise-and-interference (SINR) ratio denoted as (η) used for subsequently determining the data transmission rate over the ad-hoc links, that is:

$$\eta = \frac{\rho[r(\hat{L}_n/r - \hat{L}_n)]^{-\lambda}|h^2|}{\hat{I}_A}, \quad (2)$$

Where " $\hat{I}_A$ " and " λ " is interference plus noise and path loss exponent in a wireless channel between the needy nodes (TVs, passengers, and pedestrians) and the donor vehicular fog node (SVs) over the ad-hoc link with values in Table 2. " ρ " is transmission power by the nodes considered to be part of the infrastructure. And " $\hat{L}_n$ " and " r " are the current location of vehicles and radius of the controller node region as shown in Fig. 2 and discussed in Section 3.1.1. Furthermore, the uplink channel is represented by the Rayleigh fading channel, denoted by " $|h^2|$ " [45]. Using the " η ", the transmission data rate on a wireless channel between the needy node and vehicular donor fog node (SV) denoted as " σ " over the ad-hoc link is possible as shown in the following equation.

$$\sigma = \hat{B} \log_2(1 + \eta), \quad (3)$$

where " $\hat{B}$ " is the bandwidth of a wireless channel between the "needy" node and the vehicular donor fog node (SV).

3.3. Time-response during offloading

Using the transmission data rates, we used separate mathematical expressions to predict how long it would take for the donor node to respond after a node in need makes a request. The donor node can be from an infrastructure like RSU, a local data center, or a cloud server using a cellular link or it can be a resource-rich vehicular fog node using an ad-hoc link, thus there are mainly two different types of responses. However, the cloud server as a donor in infrastructure is considered separately due to its uniqueness in terms of physical distance from needy nodes. While another type of time response can be from donor nodes that are either RSU, local data center, or a vehicular donor fog node. A summary of each type follows.

3.3.1. Response-time-estimation for RSU/SV/local data center

We combined the estimating time to offload to donor RSU or local data center designated as " α " and donor fog-vehicle (SV) " β " as illustrated in Eqs. (4) and (5), respectively and obtained " γ " as in Eq. (6). Because RSUs and data centers are typically thought to be in the same physical location relative to the recipient node, we used the same time estimation model (α) for both types of donor nodes and a different time estimation model (β) for the vehicular type of donor node. We combined two equations Eqs. (4) and (5) to utilize both types of transmission data rates eventually due to the difference in the two wireless channels.

$$\alpha = \frac{\hat{S}_z}{\hat{R}}, \quad (4)$$

$$\beta = \frac{\hat{S}_z}{\sigma}, \quad (5)$$

$$\gamma = \sum_{V_N} \sum_{T_N} \sum_{G_\theta} [2 \times \alpha + 2 \times \beta + G_\theta[\zeta + \delta]], \quad (6)$$

where “ G_θ ” is a variable that we used to designate offloading decisions among cloud servers, fog vehicles, RSUs, and local data center. “ ζ ” and “ δ ” represent the time spent by tasks waiting for the donor-fog vehicle or RSU to execute them and the time spent by the donor-fog vehicle or RSU during execution, respectively.

The dependence on the data rate of the type of application has also a large influence on data-transmission rates. In this context, the 3GPP recommendations helped us identify even the most advanced vehicular applications along with basic vehicular applications and a wide range of applications [15]. This identification of different applications helped us to finalize the much-needed data size as its values are shown in Table 2. The Data-size “ $\tilde{S}_z$ ” is one of the requirements that impact the response time as can be seen in Eqs. (4) and (5). The 3GPP release-16 standard recommends establishing the data size of a single packet to be a function of higher layers [15]. The 3GPP release-15 and its later releases helped us to name this fixed data size as a transport block (TB) with value as in Table 2.

3.3.2. Modeling time-to-offload to the cloud-server

The powerful multi-core cloud servers are the fallback plan for many computation-offloading strategies that rely on fog nodes and edge servers. Also, it is a highly optimistic idea, to expect in time augmentation of latency-sensitive computation tasks offloading process by cloud server especially because of the high communication delays while traversing of task from need node to cloud server. Despite the fact of difficulty to achieve a successful latency-sensitive offloading model, with the use of condition-02 of our optimization problem, as described in Section 4, we modeled the cloud server response time to achieve a time response of less than 100 ms. However, to get the response time specifically from the cloud server, we used the expression for determining the cloud-server response time as:

$$\widetilde{T}_C = \frac{\tilde{S}_z}{B_C} + t_C, \quad (7)$$

In Eq. (8), t_C represents the time it takes for a cloud server to complete a given task and “ $\widetilde{B}_C$ ” is the bandwidth as defined in Table 2. The processing power of a cloud server as shown in Table 2 is huge, and is used in the denominator in Eq. (8). “ t_C ” though is a negligibly small value, but is included in our model as it represents a relevant assessment parameter.

$$t_C = \frac{\chi}{p_C}, \quad (8)$$

An additional critical aspect is the backhaul network’s impact on the cloud server’s overall response time. In order to get to the cloud, data from the client end typically has to travel across a Backhaul network. Following is how we figured out how long it took data to go over the backhaul.

Backhaul network’s elapsed time estimation and complexity. The backhaul network, an intermediary element of the network between needy nodes and cloud servers in our VFC scenario in the Fig. 1, has a complex network structure. The need for massive bandwidth between the consumer and the cloud server, the structure of backhaul networks, and the associated delays are a few of the key reasons for such complexity. Therefore, we dealt with such multiplicity by employing channel bandwidth between RSU and cloud-server as in Eq. (7) for calculating the transmission rate between RSU and cloud-server, symbolized as “ $\widetilde{B}_C$ ”. We obtained the idea of using data rate and bandwidth as in Eq. (7) from the article of similar work published as [39]. However, backhaul networks are an essential component. Moreover, exploring additional ways other than bandwidth and data size for predicting traversal time across backhaul networks are critical for developing new frameworks and analysis of existing models.

3.3.3. Time estimation during RSU/SV-CU/TV task feedback (or downlink latency model)

In addition to estimating uplink communication time from needy nodes to donor nodes, the total-response-time model should be able to estimate downlink communication time, which is the time spent conveying data from donor nodes to needy nodes. Several computation-offloading frameworks based on total response time have neglected how down-link latency impacts the system; for example, see the article referenced as [39]. These frameworks eliminated the use of downlink delay for various reasons, which include maintaining things as simple as possible in the model. Nonetheless, omitting the “downlink” delay have a significant impact on the real value computation of total reaction time. Furthermore, for an effective offloading system, the “downlink” latency must be at least equal to the “uplink” latency. However, a significant topic is how to more correctly predict uplink and downlink latencies. In this regard, the paper [16] suggests that obtaining sufficient accuracy is dependent on choosing appropriate task scheduling methods and data packet size. However, in all instances, and given the importance of downlink channel modeling, our present VFC framework depends on response time estimation based on the uplink channel. We utilized the same model for both the uplink and the downlink in our current VFC model.

3.4. Computation-execution time model

The next important phase after determining the possible response time is determining the time consumed by task execution after tasks are offloaded to potentially feasible donor nodes. With the presence of smart vehicles and RSU in the same neighborhood, we used similar mathematical expressions for calculating computation processing power, denoted as “ δ ” as in the following Eq. (9).

$$\delta = \frac{\chi}{p_D} \quad (9)$$

Where “ p_D ” denotes the processing capability of SVs or RSUs, “ χ ” denotes the amount of processing power needed to complete each individual task, with values falling within the range shown in Table 2. The computation processing power other than that of the smart vehicles and RSU is that of the data-center and cloud server. We grouped the computation processing power of data center and cloud-server together and considered mathematical expression in Eq. (8) for determining their task execution power. The “ χ ” values use in the above Eq. (9) in Table 2 are only the beginning; in our upcoming models, we will use the capabilities of the most contemporary OBCU for autonomous vehicles, such as Qualcomm’s QCM2150 System-on-Chip (SoC). Some enterprises are leading the way to develop highly enhanced computing SoCs for autonomous vehicles. According to the report referred to in [46], NVidia’s Graphics Processor Unit (GPU) SoC is one prominent example that will be integrated into Mercedes’ autonomous vehicles by 2024. Winding up the discussion with a relevant question, why did we use the values for “ χ ” other than those of contemporary RSUs and OBCUs? The answer is that we followed values in a published article cited as [39].

3.5. Waiting time by tasks at RSUs and SVs

Incoming tasks may be delayed owing to waiting time even with advanced OBCUs for RSUs and SVs and communication protocols [47]. Many computation-offloading frameworks have shown the effectiveness of including “waiting time” in the design of scheduling methods. In one computation offloading method cited in [47], an asynchronous slave-master strategy is used to cut down on wait times. Thus, the final optimization problem formulation of VFC frameworks still requires a precise waiting time model, even when including systems based on the most recent NR-V2X or IEEE 802.11bd, as in one publication cited as [16]. No method, not even the most optimized like semi-persistent scheduling, can guarantee that all waiting times will be eliminated [48]. In semi-persistent scheduling, a controlling node allocates dynamic resources to user equipment (UE). So, how does our “waiting time” model stack up against those used by competing offloading architectures? Instead of explicitly adding it in the optimization process, “T”, as in one of the latency-optimized VFC frameworks provided frameworks in the article given as [39], we modified “waiting time” symbolized as “ ζ ” by segmenting our framework into discrete time-periods. The time segmentation greatly decreased the total waiting time, especially for those tasks that must wait in line before being completed. Another reason is that the appropriate use of V2X paradigms for offloading, as described in Section 3.1.3, effectively distributes the burden due to tasks’ waiting at distinct entry points to any of the four external computation resources, namely Donor vehicular fog node, RSU, local data center, and a cloud-server.

4. Utility function

In this section, a utility function is developed for offloading to smart vehicles, RSU, data center, or cloud server. It ensures the needy node will receive back timely executed tasks after offloading to potentially feasible donor nodes. We use the following factors in the construction of the utility function.

- Communication delay:

A crucial factor in determining a good offloading decision by a potential controller node is traversal time, or communication delay consumed by tasks from needy to donor nodes. The whole TSM process relies on the status updates of all the potential and feasible donor nodes to calculate this important metric, which is the communication delay. Therefore, based on the received status updates regarding the expected consumed time during communication is determined using Eqs. (6) and (7).

- Task execution delay:

A key statistic for each decision that must be made instantly is the time taken to complete the task’s computation by a particular donor node as depicted by mathematical expressions in Eqs. (8) and (9). As the processing time between operations decreases, the offloading process becomes more reliable.

- Waiting delay:

With multiple antennas at each node, it seems that waiting at a potential donor node by an incoming offloading task may be negligible. However, studies suggest if additional tasks were assigned to the same vehicle before the current period or the waiting time is longer than intended, the offloading decision would be affected. Also, vehicles that stay within range for an extended period of time may have tasks piling up while those with more redundant computation risk losing contact. So, based on these factors, the utility function should contain waiting time as one of its components.

Using the above-mentioned factors, the entire utility function takes the following mathematical form.

$$T_{gr} = \sum_{V_n} \sum_{Tks} G_{\theta} \times \gamma + (1 - G_{\theta}) \times \widetilde{T}_C, \quad (10)$$

After finalizing the assessment models to estimate the data transmission rates of the wireless channel, computing execution time by the donor's external computation resources, and waiting time at entry points at donor nodes, a binary offloading-based offloading optimization problem with " G_θ " as the main decision variable is formulated in Eq. (11).

$$\begin{aligned}
& \underset{G_\theta}{\text{minimize}} \quad T'_{gr} \\
& \text{subject to} \\
& C1 : \sum_{Vn} \sum_{Tks} I_{FR} = S_{FR}, \\
& C2 : \gamma \leq C_\tau, \\
& C3 : G_\theta, SV_\phi \in \{0, 1\}, \\
& C4 : \sum_{l=1}^{u=N_T} SV_\phi = 1, \\
& C5 : \sum_{[1,T]} P_T + P_{T-1} \leq FR_A, \\
& C6 : T'_{gr} \leq \gamma \leq T_H \\
& \quad \quad \quad \% \text{Helps SVs or RSU} \leftarrow Tks \\
& C7 : T'_{gr} \leq \widetilde{T}_C \leq T_H \\
& \quad \quad \quad \% \text{Helps Data - Center/Cloud} \leftarrow Tks
\end{aligned} \tag{11}$$

We adopted binary offloading because it allows for a straightforward design at the expense of some amount of processing power, which we first assumed existed in potential controller nodes. In conclusion, the form of our final optimization problem is formulated in Eq. (11) where constraints C1 to C7 are briefly described in the following text.

C1: RSU's and donor-fog vehicles' maximum strength limit to handle the number of incoming requests in a given time, " T ", " S_{FR} ", should be used to determine the maximum number of requests for task offloading, " I_{FR} ".

C2: The total response time " γ ", by the donor RSU, fog-vehicle (SVs), data center, or even a cloud server to the needy CU/TV, as shown in Eq. (4), should be smaller than or at least equal to the connection duration between these two nodes, symbolized as " C_τ ".

C3: By imposing this restriction, we secure that both the first and second phases will only involve binary offloading, rather than partial offloading.

C4: This condition is specific to donor vehicular fog nodes. After assigning a task to a single SV (fog vehicle), no additional SVs should be used until the assignment has been finished (SV).

C5: If the summation of required computation power of tasks in the time period " T " " χ " and in the previous time period " $T-1$ " " T_{pC} " is more than " FR_A ", the RSU (controller) will not allocate tasks to the fog-vehicle or RSU. The symbol " FR_A " stands for the available computational resources of fog vehicle (SV) or RSU during " T ".

C6-C7: These conditions explicitly ensure the scheduling and allocation of the external devices to tasks' requests in any time period, " T ". One of the prime factors of dealing with the worst-case scenario is the presence of a tolerable threshold latency " TH " as in Table 2, and other response latencies that are calculated using the equations mentioned before this optimization problem formulation. The " TH " helps in aiding leniency to reduce the number of failures by allowing another overall latency limit as an acceptable upper bound with value as in Table 2. The conditions associated with scheduling and allocation to any of the four resources are significantly clear from their constraints' expressions.

5. Temporal segmentation and modular (TSM) based distributed approach for scheduling and allocation

In this section, we present a TSM framework for offloading processing in a VFC network with the objective to timely complete delay-sensitive, computationally intensive tasks. The timely execution of delay-sensitive tasks relies on optimal offloading and allocation of computational resources. Therefore, after scheduling computing resources, the offloading strategy is finalized in this scheme, and finally, resource allocation is developed in the context of a particular offloading approach. In addition, utility functions are the most important factors to consider while making decisions in order to reduce overall response latency, which is achieved by employing two unique modules as shown in Fig. 3. The following is a breakdown of the strategy and its modules in detail:

5.1. Module 1: First-phase decision for task-offloading model to SVs, RSUs, or cloud-server

Before we get into the specific parts of this framework, it's important to note that the question of whether or not to outsource a given task is likewise a hotbed of research activity, and so need to be included in the beginning of any task offloading technique. However, we are considering TSM with a focus on optimizing offloading processes once the consumer decides whether to offloading the tasks to a computation-rich resource. Therefore, we used a simplex approach that employs a random selection to make decisions for the consumer. Further, after the consumer end decision to outsource to any of the external computation resources, the next step is a good decision to select and schedule the potential and feasible donor nodes. A reliable computation resources selection process should encompass multiple dimensions to leverage a VFC-based distributed computing scenario effectively. Keeping this fact

in mind, our first algorithm encompasses this module and is based on resource availability and an estimated total response time by donor nodes using the definitions in the utility function in Eq. (10) and optimized in Eq. (11).

Algorithm 1: Selection between donor fog nodes, RSU, data center, or Cloud-server

Data: Algo-1 at the donor node gets consumer requests for task offloading

Condition: Algo-1 controller nodes may be RSU or fog nodes

```

while All the tasks during time-period "T" do
  if Donor-node resources are available then
    if Approached Donor-node meets the criteria in Eq. (11) then
      if Donor node is RSU then
        if Type of needy node is vehicular fog-node then
          Step 1: V2I criteria applies
          Step-2: Approached Donor  $\leftarrow$  tasks
        else
          Step 1: V2P criteria applies
          Step-2: RSU  $\leftarrow$  tasks
      else
        if Donor node is a vehicular fog-node then
          if Type of needy node is pedestrian then
            Step-1: V2P criteria apply to tasks offloading
            Step-2: Approached Donor  $\leftarrow$  tasks
          if Type of needy node is vehicular fog-node then
            Step-1: V2V criteria apply to tasks offloading
            Step-2: Approached Donor  $\leftarrow$  tasks
    else
      if Neighborhood SVs has resources and meets Eq. (11) criteria then
        Step 1: V2V or V2P criteria applies
        Step 2: Invoke Algorithm 2 else
          if Possible to allocate tasks to the neighborhood data center then
            Step-1: V2I or P2I criteria apply to tasks offloading
            Step-2: Data-Center  $\leftarrow$  tasks
          else
            Step-3: Cloud-Server  $\leftarrow$  tasks
  
```

5.1.1. First algorithm scheduling of tasks to SVs, RSUs, the data center, or the cloud

The pseudocode in Algorithm 1 is based on resource-availability and estimated total response time by potential donor nodes. In this step, we followed mode-1 recommendations in SideLink (SL) of the 5G NR-V2X. According to mode-1 recommendations, the potential controller node, allocates resources to donor nodes based on Workflow as in Fig. 3. The connection to 5G Radio Access Network (RAN) via "gNodeB" is shown in Fig. 1. The 5G-RAN keeps the connectivity of RSU, TVs, passengers and pedestrians with potential donor nodes. The Algorithm 1 keeps track of one time-period prior status updates as can be seen in red colored workflow diagram in Fig. 3. These updates include probable resource availability, time consumption during communication, computation execution time by donor after receiving the task, traversal time after execution back to needy node, and waiting time in case of any bottlenecks and queuing delays because of computation resource unavailability on a scheduled donor node. In Eq. (10), we aggregated these time consumption as a utility function.

In the next step, as shown in Workflow Fig. 3, needy nodes can invoke several potential controller nodes in the neighborhood during any time period, "T." As TSM follows the distributed scheduling and allocation procedure, any approached donor node by a needy node can be a potential controller node. The potential controller node upon receiving the task request, runs Algorithm 1 and decides in favor of itself, or among the other four potential donors. Here resource availability and expected time consumption are the two primary factors that influence this key decision. If the Algorithm 1 decides in favor of itself, it directly assigns the task for execution but if this decision is in favor of other resources then it uses a specific decision variable, symbolized as " G_θ ". If the decision is in favor of the value of donor fog vehicles (SVs) then it is indicated by a value "0". But if the decision is in favor of the other three donor nodes, then it is indicated by a value "1". Among the other three donor nodes are "RSU", data center, and cloud, and these are preferred in the same order as shown in TSM workflow in Fig. 3. If the decision among any of the donor nodes is in favor of donor fog vehicles (SVs), then Algorithm 2 (explained in the following text) needs to be run to find the appropriate SV

Algorithm 2: Identifying the most feasible donor vehicle fog node**Input:** Donor Vehicular Fog-node receives tasks**Data:** Save reports on neighboring donor nodes status in "SVs-array"**Step-1:** Sorting (descending) incoming tasks by computational power**Step-2:** Sorting SVs by computational power (ascending)**Step-3:** Save size of "SVs-array"**while** Maximum tasks per time period, " T " **do** **if** Tasks from another donor node and satisfy Eq. (11) **then** **if** needy node is pedestrian **then**

V2P criteria applies

if needy node is TV **then**

V2V criteria applies

Step 2: $SV \leftarrow$ Tasks **else** **If** Tasks are received from Needy node **if** needy node is pedestrian **then**

V2P criteria applies

if needy node is TV **then**

V2V criteria applies

SVs-array-copy1 = SVs-array;

index1 = 1;

while size of SVs-array $\geq$ size of SVs-array-copy1 **do** **if** Average response time (utilization) according to Eq. (11) of SVs-array-copy1(index1) $\leq$ that of SVs-array-copy1(index1+1) **then**

index1 = index1+1;

else

Pop out SVs-array-copy1(index1) array entry

Step 3: % Record keeping...

SVs-array = SVs-array-copy1;

return SVs - array % that is sorted % SVs in descending order

% with respect to maximum

% computation power and least

% average response time

if index1 == size of SVs-array **then**

Stop the inner while loop

Step 4: Final allocation **Result:** SVs-array(index1) $\leftarrow$ Top-sorted task**5.2. Module 2: The second phase of intra-fog-vehicle (SV) scheduling**

The Algorithm 2 employs the guidelines of all the sub-modes of mode-2 of SL NR-V2X. These sub-modes range from mode 2(a), 2(b), 2(c), and 2(d). In contrast to SL mode-1, NR-V2X mode-2 provides four sub-modes, none of which are dependent on 5G-RAN [19]. Outlining the guidelines of all these modes one by one, through a sensing-based semi-persistent transmission strategy, each UE in 2(a) may make its own resource selection independently and normally endorses V2V connections. However, resource selection in our case is the task of the potential controller node, and TSM designates and fix the first approached donor node as the potential controller node for scheduling and allocation tasks. Next is decentralized scheduling approach 2(b) which allows UEs to work together to choose the most efficient means of data transfer specifically from Vehicle to Infrastructure (RSU and data center in our case). Mode-2(c) guidelines rely on total-response time by SVs, as indicated in Section 3.3.1 and is more specific for V2P type of interactions in case the needy node is pedestrian. Finally, a UE may plan the SL transmission for other neighboring UEs in mode-2(d) and is designated for V2N class.

Similar to Algorithm 1 in the first step, we modeled our second step using resource (fog-vehicle) availability and total time-response-based scheduling, in which a controller node selects the fog-vehicle based on the shortest response time (SVs). Pseudocode in Algorithm 2 depicts the main components of the second-phase scheduling process, including the retrieval of requests for tasks at step 1, the sorting of tasks according to their computation power in descending order, and the sorting of vehicle computation power in ascending order at step 2, record keeping at step 3, allocating maximum computation-resources to a fog-vehicle with minimum available computation resource at step 4, and updating table in step 3 again at step 5. The dual-sorting process serves to pair high-resource-demanding tasks with vehicles that have fewer computational resources. Priority is given to the task that necessitates the most computational resources. To prevent the under-use of redundant computing resources of SVs, priority is given to vehicles with the lowest computational requirements. Therefore, the key factor on which 2nd-phase scheduling depends is the total-response time which is the amount of time it takes for donor nodes (fog vehicles) to process and finish a request to undertake a task.

5.3. Numerical results

This section discusses our method and presents numerical results. We assume an RSU with a VEC server, a local data center, smart automobiles along considered roadways, and a cloud server in an urban area. Each vehicle's computational resource is assumed to be [10, 15] GHz in our scenario. We used the vehicle's density to be [50, 100], computation processing power required by different tasks [50, 100] cycles per second, wireless channels interference and noise as $[-70, -75]$ dB, path loss component of 3.2, transmission power of smart vehicles, RSU and data center in the range [80, 120] mW and a maximum latency threshold T_H set to 250 ms. We further confined the range of our defined ad hoc-links to 150 m and the range of cellular links to 200 m. We presume that the potential controller node broadcasts beacon messages containing computation resource information to the donor and needy nodes within its communication range. Once a link is established, communication between the potential controller node, donor, and needy nodes switches to unicast mode. Additionally, all donor and needy nodes periodically share their pertinent information with the potential controller node. We selected a 5G NR-V2X connection in our case since the message sizes are small relative to the greater bandwidth used in 5G NR-V2X [41], therefore, we overlook communication overhead.

Our simulation environment relied on MATLAB (Release 2022a (R2022a)) running on an Intel(R) Core(TM) i7 13700F with 32 GB of RAM. During each given T -second simulation period, there are V_N vehicles and T_N requests for the completion of various tasks. All the input parameters are included in Table 2. The proposed TSM method is compared to the baseline methods illustrated below:

- **Round Robin (RR)** A round-robin is a deterministic scheduler that rotates around each connected device or vehicle in turn, giving each one a share of the available resources. Deterministic scheduling and allocation strategies in 5G NR (New Radio) are those that allocate resources in a consistent, predictable manner according to a set of rules and priorities. These procedures are neither heuristic nor probabilistic, but rather methodical and predictable in their approach to allocating resources. The goal of deterministic scheduling is to assure service availability and performance for all customers and services. In other words, the RR scheduler begins by allocating a block of resources to a single device (fog vehicle, RSU, data center, or a cloud), and then proceeds to the next device or vehicle in the queue until all devices or vehicles have been allocated their share of the available resources. After then, the procedure resumes from the very beginning.

With this method, all devices, and vehicles, regardless of their importance or quality of service needs, have an equal chance of accessing the available radio resources. However, this might result in lengthier wait times for devices or vehicles that have greater priority or more stringent QoS requirements.

RR may not be the option to use in every situation, but guarantees a reasonable allocation of resources in conditions, for example, VFC-enabled systems with:

- Relatively increased simplicity and lower complexity,
- Requirement of increased fairness in resource allocation and scheduling,
- Need of small workload or number of users,
- Short data transmission intervals,
- Low variations,
- Low latency requirements, and
- Small cell deployments.

- **Proportional Fairness (PF)** This computation offloading proportional fairness scheduling & allocation method based on 5G NR (New Radio) V2X (Vehicle-to-Everything) is a heuristic approach employed in several recently published articles to provide a fair distribution of resources across various V2X devices [49]. Major features of a PF approach include fairness based on QoE, efficiency improvement by optimizing total system capacity and spectrum efficiency, quality of service (QoS) differentiation to provide sufficient QoS levels for all devices, and flexibility to change network environments. On the other hand, PF methods include limitations such as complexity, a lack of optimal performance due to a trade-off between fairness and efficiency, increased latency and delay, and susceptibility to inaccurate channel estimates. In Section 5.3, we discuss our findings regarding the efficacy of this strategy in the simulated scenario, taking into account its pros and cons by comparing it with other potential methods.

Proportional Fairness (PF) is a 5G NR-based scheduling technique that balances system performance and user resource distribution. PF is better in certain situations:

- Diversity in consumer needs,
- Changing channel parameters,
- Flexible allocation of resources,
- Optimal Use of Available Means,
- Consumers prioritization,
- Instantaneous support, and
- Balancing Workloads

- **Best-Channel Quality Index (CQI) Scheduling and Allocating**

This technique can be thought of as a subset of proportional fairness (PF) approaches. PF is considered a superset of the BCQI method because the PF method utilizes multiple factors to make appropriate decisions in the scheduling and allocation

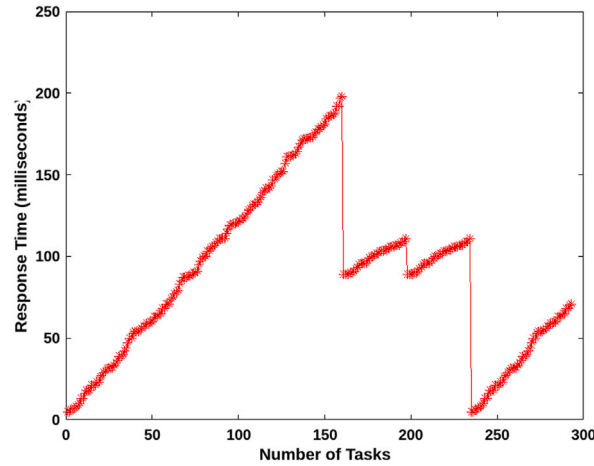


Fig. 5. TSM Scheduler curve for response time versus number of tasks.

of resources. Such decision criteria may include the data rate, latency, QoS, QoE, and Channel Quality Index (CQI). In the BCQI, the CQI determines balanced scheduling and allocation [50]. However, there are multiple implementations that are termed by the developers as BCQI scheduling and allocation i.e., “Dynamic CQI-Based Scheduling”, “Quality-of-Service (QoS)-Aware Allocation”, “Adaptive Allocation”, “Multi-User Diversity”, and “Hybrid Approaches” are some examples. These implementations mostly stray from the core notion of BCQI scheduling and allocation, where the sole affecting element is BCQI. We, therefore, in our current experimentation stage, tested the type of definition that relies on a single factor of BCQI for prioritizing tasks and their final allocation to potential donor devices. However, experimenting with more variations of the procedure yields greater fairness in determining the merits of methods in comparison.

The BCQI, like other methods, has pros and cons. Improved performance, resource efficiency, adaptation to channel conditions, and QoS differentiation are a few of its main features. Whereas, complexity, channel estimate inaccuracy, fairness challenges, delay, and overhead can be listed as its limitations [51,52]. In the discussion Section 5.3, after evaluating the pros and cons of this approach, we explain our results based on its performance in the simulated scenario. Lastly, as mentioned, ‘BCQI’ is a subset of ‘PF’, and the situations in which ‘BCQI’ is more suitable are the same as those of the ‘PF’ method.

Our analysis depends on the relationship between response time and a number of incoming tasks with a specific method in use as a scheduling and allocation method which is validated by running the VFC-scenario as shown in Fig. 1 each hundred times with specific scheduling and allocation method. As a result, we obtained probable response latency ranges with respect to incoming tasks requests for the scheduling and allocation methods as shown in Fig. 7. The Monte Carlo analysis-based Probability Density Function (PDF) (Fig. 8) of the obtained relationship between response latency and a number of incoming tasks, helped us verify the response latency range by our considered scheduling and allocation methods. We have avoided placing Fig. 8 here and instead have shown it near Section 5.3.1 as more relevant details regarding Monte Carlo analysis-based Probability Density Function (PDF) are mentioned in Section 5.3.1. Hence, another relationship is the run time that represents the relationship between run time and the number of incoming tasks as shown in Fig. 9. We have avoided placing Fig. 9 here and instead have shown it near Section 5.3.2 as more relevant details are mentioned in Section 5.3.2. Hence, the run-time helped us validate the behavior of a particular scheduling and allocation method to a number of incoming task requests. The following paragraph describes how we used run time and Monte Carlo analysis using PDF-graphs:

5.3.1. Response time vs number of tasks using Monte Carlo integration-based analysis

We used Monte Carlo integrals, which is a numerical method, to determine and compare the efficiencies of the methods as in Figs. 5–7 [53]. We used the numerical method to avoid the complexity of methods that rely on conventional mathematical evaluations. One of the main reasons for such complexities is the complexity of transfer functions of the considered scheduling and allocation methods. Especially the behavioral complexity of our approach and other baseline methods can be seen from the relationship graphs between response latency and incoming tasks requests in Fig. 7. But to obtain Monte Carlo analysis, we followed a specific procedure. First, we elaborate on the steps involved in performing this particular type of analysis process, these are:

1. Identifying the problem (as in Eq. (11))
2. Defining the input parameters by calculating the area under the curve (Monte Carlo Integral) using Eq. (12) for each method after running and simulating a VFC-network scenario for each method that provided us the relationship between response latency and number of incoming tasks as shown in Fig. 7.
3. Outlining distributions of probability (by running the VFC-network scenario in Matlab for hundred times for each method)

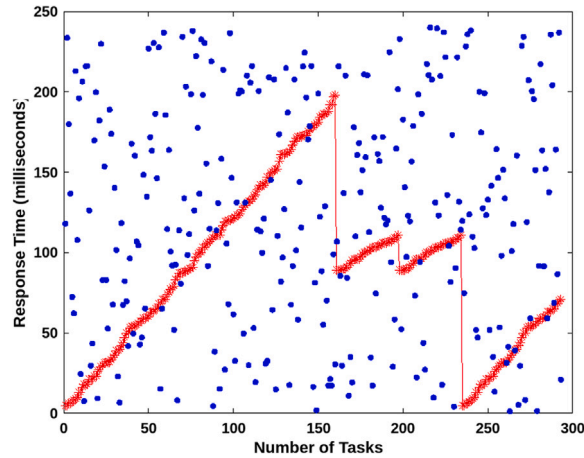


Fig. 6. Placing of the random sample points (response latency) above and below the scheduler curve of TSM scheduler.

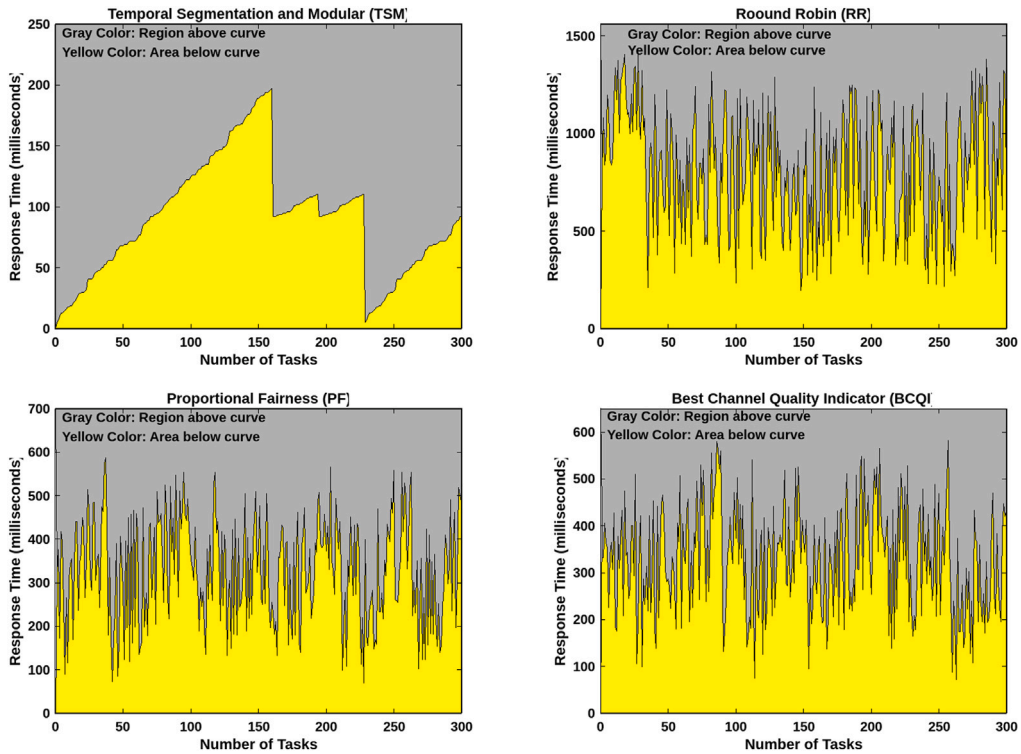


Fig. 7. Monte Carlo integration analysis based on covered area depiction of all methods.

4. Creating random samples (Based on obtained relationships between response time and incoming tasks in VFC-network scenario simulations in previous steps)
5. Conduct calculations (By finding Probability Density Function using Matlab inbuilt command “normpdf”)
6. Iterate through steps 4 and 5
7. Analysis of outcomes (using probability Density Function (PDF) [54])
8. Inspect and improve (We cross-checked the obtained relationships between response latency and incoming task requests with several results in recently relevant published literature)
9. Taking decisions
10. Reporting outcomes

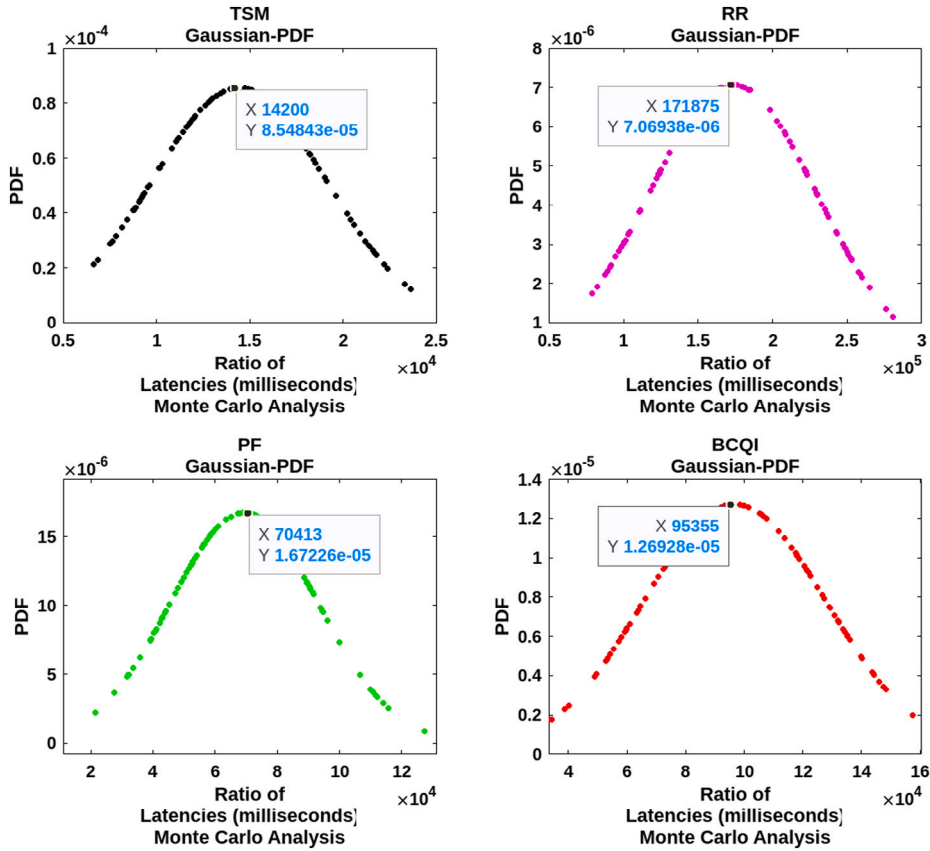


Fig. 8. Probability density function (PDF) of all scheduling and allocation methods.

Let us first go through the analysis process that helped us validate a relationship between response latency and a number of incoming tasks for our “TSM” technique. For other baseline techniques, we merely provide the final plot of the analysis process after providing the TSM method’s whole analysis process. Hence to start with, after the first step of identifying the problem using Utility-function and its optimized form in Eqs. (10) and (11), we first obtained the approximate input parameters by calculating the area under the curve (Monte Carlo Integral). This curve is the relationship between response latency and the number of incoming tasks for the TSM scheduler which is obtained by running the VFC-network scenario as shown in Fig. 5. We run the VFC-network scenario for hundred different times to obtain probability distributions of Monte Carlo Integral for the TSM curve. In each run, we placed random sample points that are randomly distributed above and below the “TSMschedulers” graph as shown in Fig. 6. After we placed the random samples above and below the graph, we then calculated the number of sample points below the graph and multiplied it by the selected (graph) area according to Eq. (12). After determining the Monte Carlo ratio, We used the same values of parameters to highlight the area under and above the curve of “TSMschedulers” as shown in Fig. 7. We calculated the ratio “ M_{CR} ” as:

$$M_{CR} = (U/N) * A, \quad (12)$$

where “U” is the number of samples with response latency below the curve of “TSM scheduler”, “N” is the total number of sample points (response latencies), and “A” is the selected area of the graph where random samples are considered.

We then considered all these hundred ratios as a random variable and calculated their Gaussian Probability Density Function (PDF). Separate distribution plots of the estimated PDFs for each approach are shown in Fig. 8. We then analyzed the range of probability distribution along the horizontal axis in the Gaussian PDF, as shown in Fig. 8 according to the stated response latencies in the figures used for Monte Carlo analysis (Fig. 7). In the case of “TSM”, the “ M_{CR} ” with maximum probability is about “14200”, whereas it is “171875”, “70413”, and “95255” for “RR”, “PF”, and “BCQI” respectively. When we calculated the corresponding percentages of the maximum probability “ M_{CR} ” ratios of the “TSM”, “RR”, “PF”, and “Best-CQI”; they came out to be 56.8%, 68.75%, 58.6%, and 59.5% respectively, meaning that these methods have the above mentioned “ M_{CR} ” with the probability of occurrence above fifty percent. Or in other words, it is highly likely that “TSM”, “RR”, “PF”, and “Best-CQI” operate within the response latency as mentioned in their Monte Carlo analysis graphs initially as shown in Fig. 7 respectively.

Table 3
Effect of different factors in milliseconds on the run time of the scheduling and allocation methods.

Method	TSM	RR	PF	Best-CQI
System architecture	1	1↔2 [55]	1↔2	1↔3
Features of the tasks	1↔2	1↔4	1↔2 [56]	1↔3
Network topology	1	1↔4	1	1↔2 [57]

5.3.2. Algorithmic run-time vs number of tasks' requests

Several variables, such as the methodology used, the system configured, the network conditions, and the nature of the task at hand, might affect the runtime of scheduling and allocation techniques in 5G NR V2X based compute offloading. Therefore, without knowing these specifics, it is difficult to generate an accurate runtime estimate. We considered the following factors as vital while evaluating the runtime of the scheduling and allocation procedures:

- System architecture knowledge
- Figure out the type of scheduling and allocation technique
- Assessing the features of the task
- Observe the network infrastructure
- Assess scheduling overhead costs
- Measure allocation overhead costs
- Estimate processing time delays
- Simulate or investigate the system
- Double-check and improve

Based on the above factors, we determined the run time of scheduling and allocation methods that we considered in our experimentation.

What are the significant factors of the highest impact and how we finalized each method can be deduced from its equation as shown in the following expression.

$$\kappa = \nu + \mu + \psi + \Omega, \quad (13)$$

whereas “ κ ” designates the “runtime” of the method, “ ν ” the impact due to system architecture in milliseconds on the “runtime” of a particular method, “ μ ” is the effect in milliseconds on the “runtime” of a particular method, “ ψ ” is the influence of network topology, and “ Ω ” is the effect of “overhead cost” on the runtime of a particular scheduling and allocation method. However, the question here is the authenticity of the values that we used to calculate “ κ ”, which is in Table 3 and used in our simulation.

One of the main reasons for selecting the values as in Table 3 is that these have been endorsed by in-progress standardization processes like Ultra Reliable Low Latency Communication (URLLC). For example, in our approach, “TSM”, based on the requirement of URLLC, we determined that the effect of system architecture on its run time is around 1 ms. Another instance is “RR” method in which the effect of the factor “System Architecture” of the type 5G NR V2X on the “RR” scheduling and allocation method, as indicated in article is in the range of URLLC. We know that URLLC endorses end-to-end latency of 1 to 10 ms; therefore, making it easy for us to choose a value within this range. A similar effect by 5G NR V2X system architecture is true for the “PF” method as used in [56], thereby we choose the effect to be in the range of 1 and 2 ms. But in the case of the “Best-CQI” method we choose the effect within the URLLC range but higher than that of “PF”. The higher range is because of the dependence on a relatively less number of factors in the case of “Best-CQI” as compared to that in “PF”.

Similarly, the effect of the features of the tasks depends on their latency requirements, which eventually depend on the nature and type of applications being considered. In our case, as we have considered computation-intensive applications with sub-hundred milliseconds end-to-end latency requirement. The sub-hundred-millisecond latency constraint restricted us to consider the effect of the “type of applications” to be in the range of 1 to 2 ms. In the case of the “RR” method, we found that the run time of the “RR” scheduling and allocation method gets affected in the range of 1 to 4 ms. This is the largest impact that type of application has on any scheduling and allocation method that we considered as shown in the Table 3. One of the main reasons for the largest effect is the inability of “RR” to schedule and allocate based on metrics like fairness, channel, and network conditions. The “PF” method, on the other hand, has a more robust mechanism to deal with the stringent latency requirement of the type of applications that we considered. Therefore, a lower effect between the range of 1 and 2 ms has been finalized as its impact on the “PF” method. Unlike “PF”, the influence on the “best-CQI” of the type of tasks that we considered in our current scenario, is more significant which is in the range of 1 to 3 ms. The obvious reason for the high impact of the type of application on “Best-CQI” is its dependence on a relatively small number of factors for the final scheduling and allocation process.

The effect of the “network topology” factor on our approach is about 1 ms. We selected this value based on the fact that the distributed approach of “TSM” makes it more robust to variations in network topology and associated complexities. Therefore, the effect of variations in network topology significantly reduces in the case of “TSM”. Next, if we discuss the impact of the network topology on the “RR” method which has definitely a significant impact on the “RR” method because RR is a very simpler approach that has no mechanism to deal with complexity, variance, or any other relevant factor. Therefore, we found the impact of network topology on “RR” to be significant that is in the range of 1 to 2 ms. However, the “PF” method is more robust because it contains

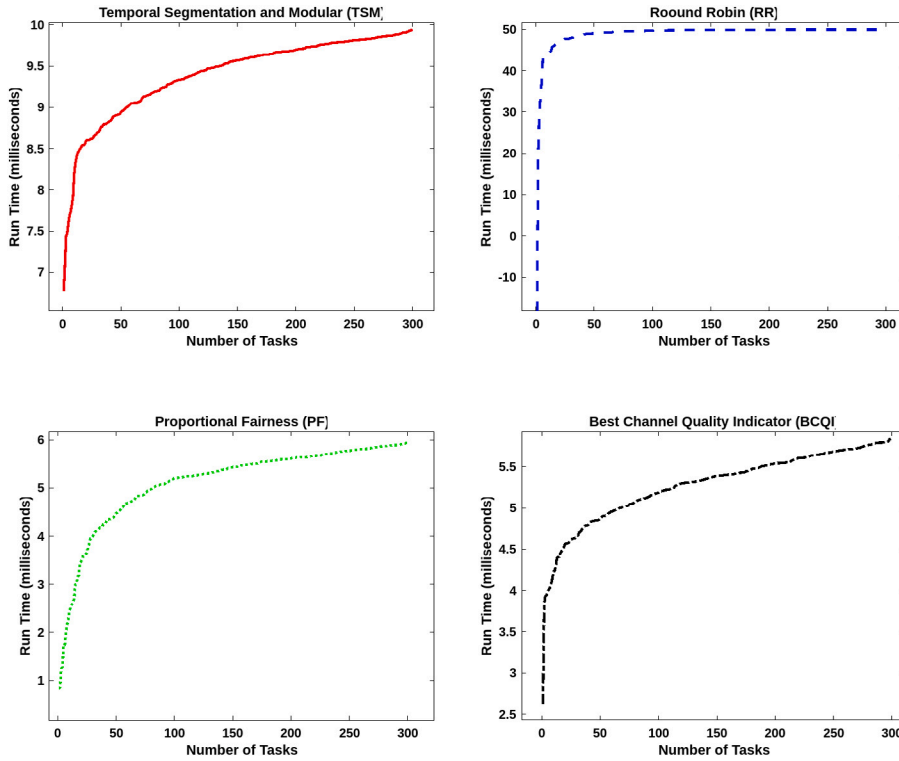


Fig. 9. Run times of all the methods.

the mechanism to tackle changes in network topology. Therefore, a relatively small effect is considered, that is, in the range of 1 ms on the run time of the “PF” method as can be seen in Table 3. Besides “PF” is the third method, “Best-CQI” can be considered as a subset of the “PF” method as it schedules and allocates the resources based on a single factor and that is CQI. The dependence of a relatively less number of factors by the “Best-CQI” method as compared to that of “PF”, makes “Best-CQI” less tolerable to changes in network topologies. This is the reason that the factor “network topology” has relatively more impact, that is, in the range of 1 to 2 ms.

As shown in Fig. 9, we compared the run times of all the approaches based on our simulation-based experiments and the results from Table 3. The results reveal that RR takes the longest running time among the four methods analyzed, followed by TSM, PF, and best-CQI. While “RR” displays the longest run-time, it also implies a nearly consistent response to an arbitrary number of task requests. The “RR” constant response may be attributed to the fact that it does not rely on any intermediary processes, such as the Channel Quality Index (CQI) or a network of channel conditions. However, a pertinent point is why “RR” exhibits long running time, given that its run time seems almost constant regardless of the number of incoming tasks. As was previously indicated, the solution lies in its seeming simplicity: it may be applied to any number of tasks without taking into account any intermediary components. The “RR” does not schedule or allocate the next inline task until after a task has been scheduled and assigned to a specific next-in-line resource. Moreover, since “RR” is fairly basic, there is a higher probability that it scheduled and assigned the current work to a resource that is already in use; as a consequence, “RR” will be waiting until the final allocation of the current task, which will prolong the duration of “RR’s” execution. Secondly, why does “RR” seem to have an almost constant “runtime”? The answer is that, though the time consumption due to no intermediate process is zero, the number of wrongly scheduled and inappropriate allocations always contributed equally to the increase in total run time, but this contribution becomes in microseconds, which seems to be almost constant. However, in computation task scheduling and allocation processes, the contribution in microseconds matters when the tolerable latency requirement is in milliseconds.

Our approach, “TSM”, has the second-highest run time and a runtime-tasks curve with a constantly rising slope. The small rise in the slope of the curve indicates that the total execution time responds very gradually to an increase in the number of task requests. This makes our method more viable than simple methods like round robin. In contrast to “TSM”, which operates in a decentralized form, other methods like “PF” and “best-CQI” have shorter run times because they rely on fewer metrics and because they function in a centralized manner. However, how much difference does the shorter execution time of “PF” and “Best-CQI” methods make in terms of the total response time of incoming tasks? Its answer can be found in the obtained relationship between total response time and incoming task requests when these methods are used as scheduling and allocation methods in a VFC-network scenario as shown in Fig. 7.

6. Conclusion

In this article, we explored a latency reduction problem where smart vehicles try to improve computational efficiency by offloading tasks in order to manage the computation resource deficiency of resource-constrained nodes. The four-type multi-tier rich computation resource devices were task offloading destinations and donor nodes. In order to optimize task offloading and computing resource allocation, we used V2X abstractions. Heuristic-dedicated algorithms were then used to solve the problem. We created a specific categorization by appropriately grouping interactions, such as cellular links and ad-hoc links, based on Radio Access Networks (RAN), between resource-constrained and resource-rich nodes in order to accomplish distributed task offloading. We created distributed algorithms and a TSM scheme that yields a nearly ideal result. The 5G NR-V2X communication paradigm was also employed to boost the system's performance. The numerical results showed that the suggested approach may decrease latency while appropriately scheduling the task requests and performing efficient resource allocation of computation-intensive applications with requirement of sub-hundred milliseconds latency. To extend our research work in future, we want to include more complicated instances where channel and vehicle status details are unclear. In order to further strengthen the task offloading process, we will investigate ways to incorporate machine learning methods by making use of 5G NR and mmWave communications to enhance long-term delay performance.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Hui Yuan reports financial support was provided by National Natural Science Foundation of China.

Data availability

https://github.com/ilyas3169/Vehicular_Fog_Computing_Latency_based_Optimization

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.iot.2023.100912>.

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